

PROJECT BUTTON HR–V0 ELECTRICAL V3 CONNECTED CANDIDATE

Separate RESET and ARM, two PNOZ stages, dual watchdog contacts, external adapters, redundant actuator interruption.

01 External listed sources and DC boundaries

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10 U2D2, actuator ports and bonding boundary

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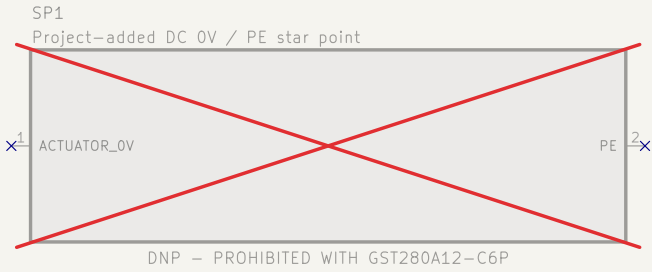
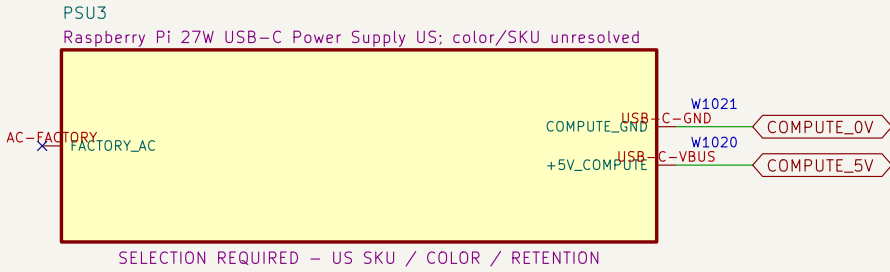
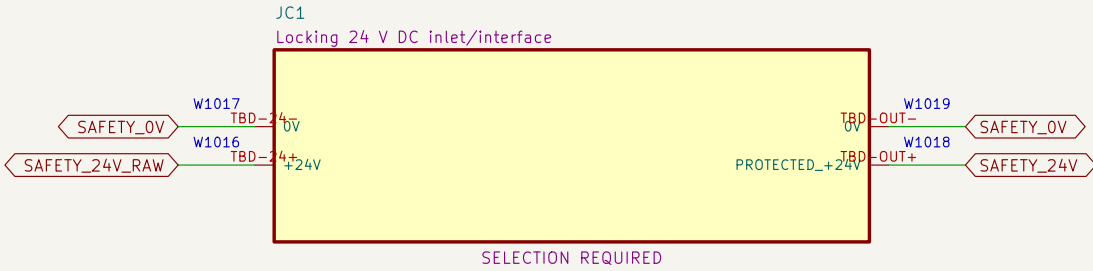
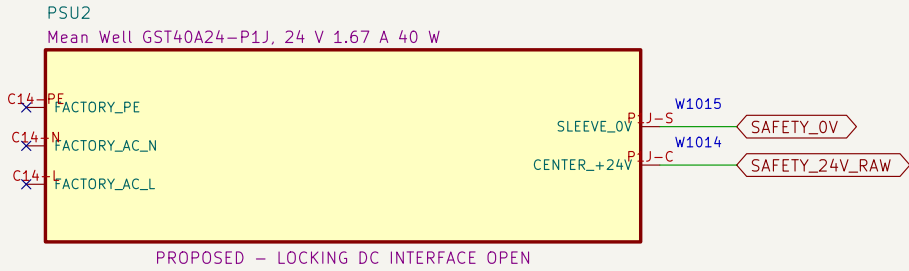
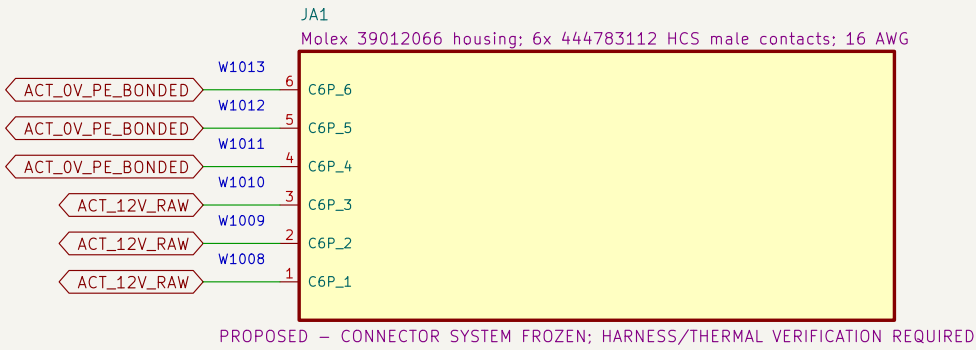
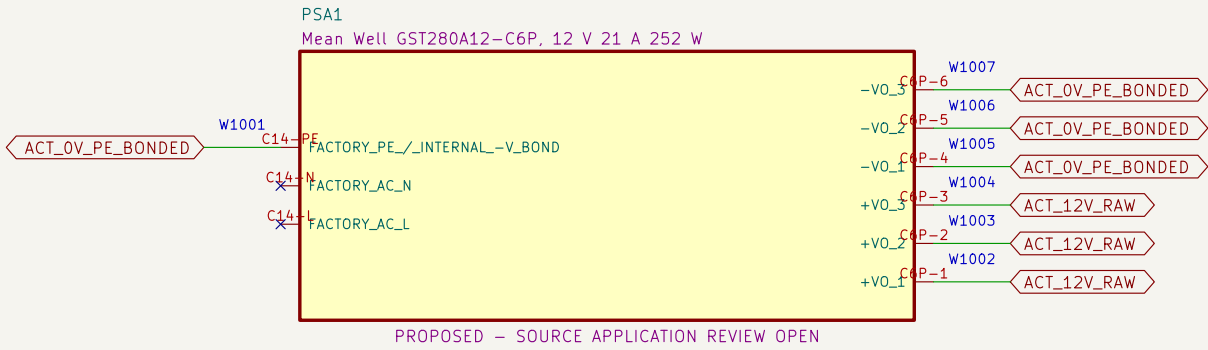
11 Watchdog PCB external connectors

File: 11_watchdog_pcb_connectors.kicad_sch

V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION Project Button		
Sheet: / File: project–button–v3.kicad_sch		
Title: Project Button HR–V0 Electrical V3 index		
Size: A3	Date: 2026–08–06	Rev: V3–P1.0
KiCad E.D.A. 10.0.5	Id: 1/12	

01 External listed sources and DC boundaries

Factory-sealed adapters eliminate project-built mains wiring.



NOTE 1: All AC conductors shown are factory boundaries, not project-built wiring.

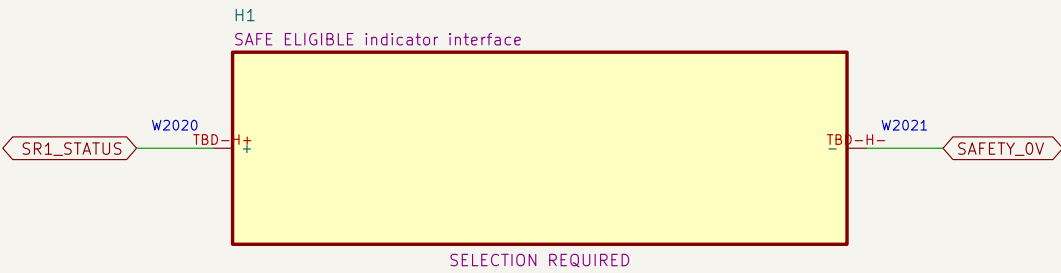
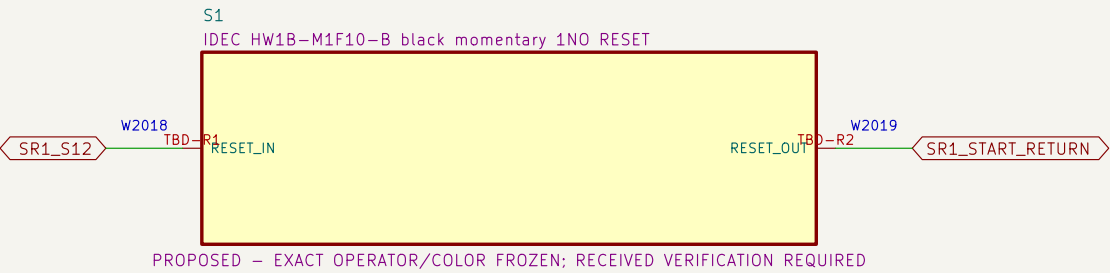
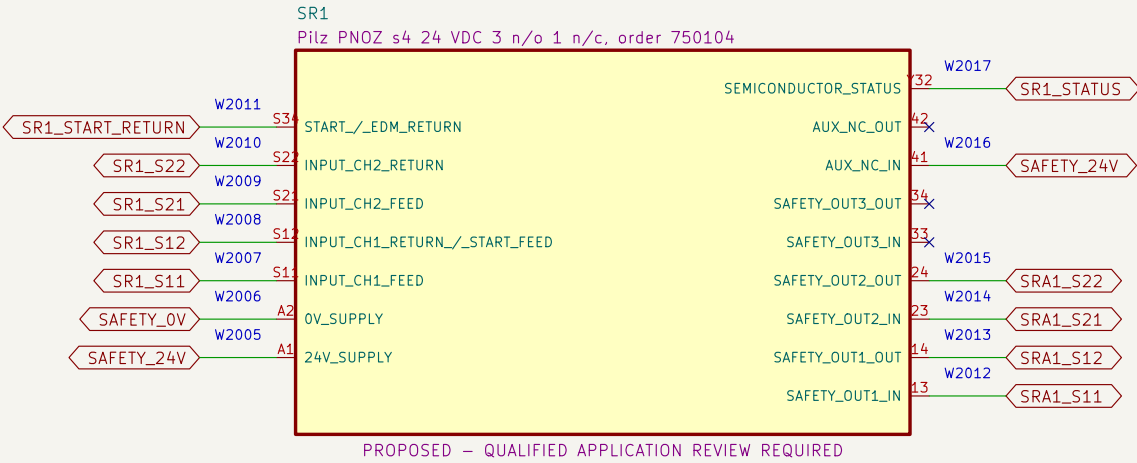
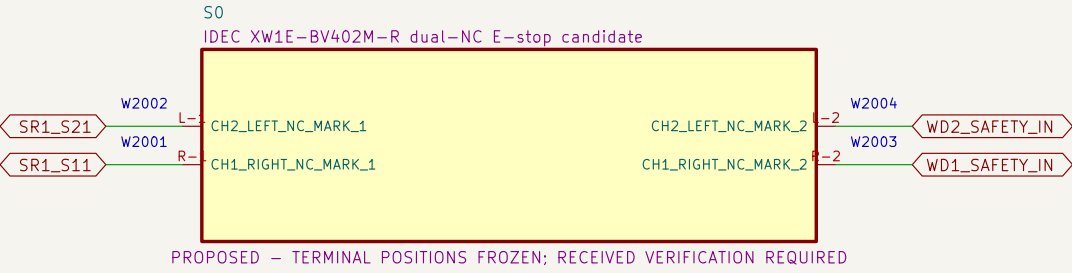
NOTE 2: Site cords, receptacles, GFCI/code basis and source application review remain open.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /01 External listed sources and DC boundaries/
File: 01_external_sources.kicad_sch

Title: PB HR–V0 Electrical V3 – 01
Size: A3 Date: 2026–08–06 Rev: V3–P1.0
KiCad E.D.A. 10.0.5 Id: 2/12

02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains one E-stop NC and one watchdog NO contact; RESET cannot energize K1/K2.



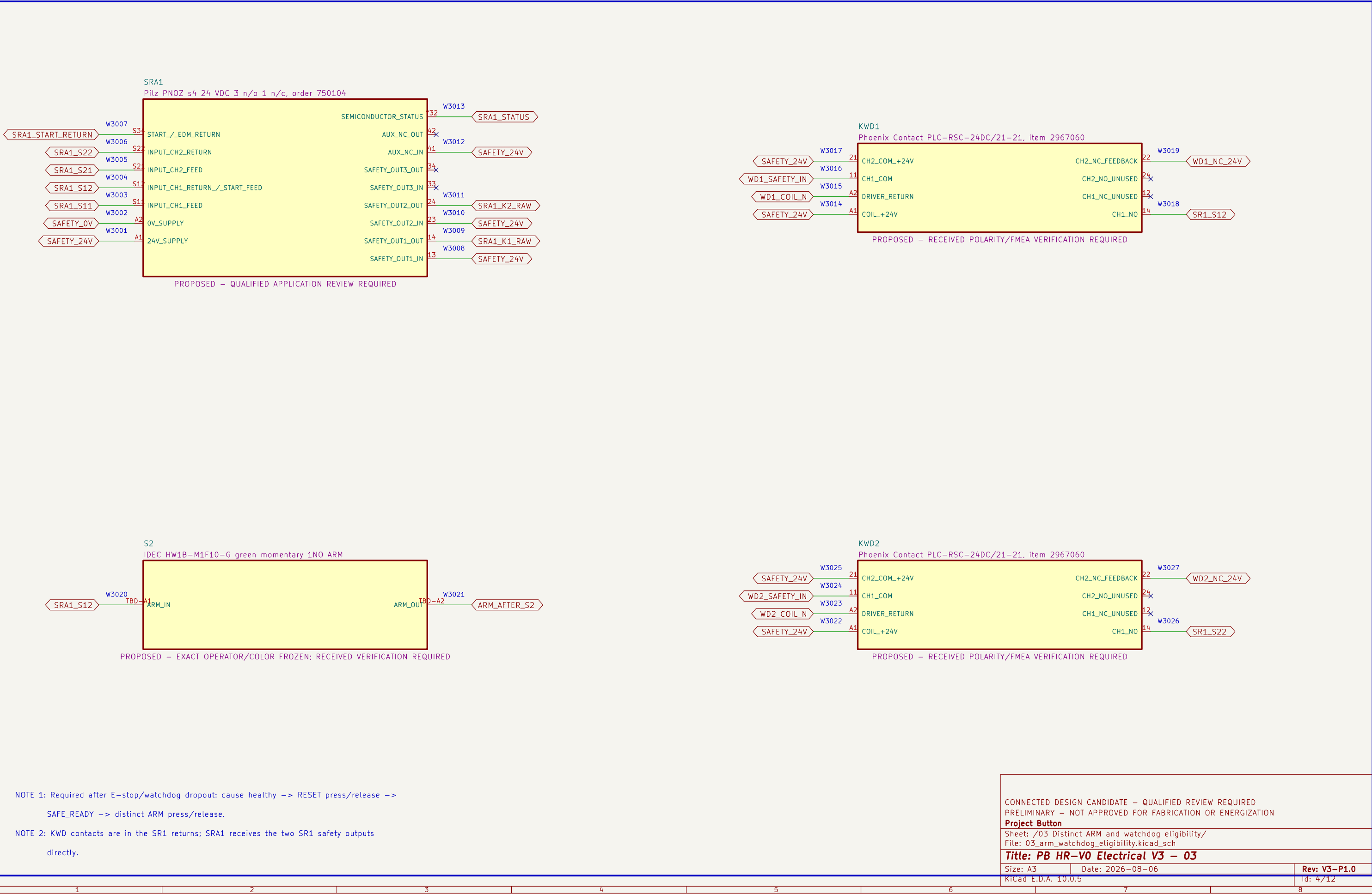
NOTE 1: Heartbeat loss opens both SR1 input returns; recovery alone cannot restore the monitored RESET stage.

NOTE 2: RESET release may make SR1 eligible, but SRA1 and K1/K2 remain de-energized until a later ARM.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /02 Dual–channel E–stop and RESET eligibility/ File: 02_estop_eligibility.kicad_sch		
Title: PB HR–V0 Electrical V3 – 02		
Size: A3	Date: 2026–08–06	Rev: V3–P1.0
KiCad E.D.A. 10.0.5		Id: 3/12

03 Distinct ARM and watchdog eligibility

SRA1 requires SR1 eligibility, two watchdog channels, EDM proof and a new ARM action.



S2

IDEC HW1B–M1F10–G green momentary 1NO ARM

SRA1_S12

W3020

TBD–A1

ARM_IN

ARM_OUT

TBD–A2

W3021

ARM_AFTER_S2

PROPOSED – EXACT OPERATOR/COLOR FROZEN; RECEIVED VERIFICATION REQUIRED

KWD1

Phoenix Contact PLC–RSC–24DC/21–21, item 2967060

SAFETY_24V

W3017

21

WD1_SAFETY_IN

W3016

11

WD1_COIL_N

W3015

A2

SAFETY_24V

W3014

A1

CH2_COM_+24V

CH1_COM

DRIVER_RETURN

COIL_+24V

CH2_NC_FEEDBACK

CH2_NO_UNUSED

CH1_NC_UNUSED

CH1_NO

22

W3019

WD1_NC_24V

24

W3018

SR1_S12

PROPOSED – RECEIVED POLARITY/FMEA VERIFICATION REQUIRED

KWD2

Phoenix Contact PLC–RSC–24DC/21–21, item 2967060

SAFETY_24V

W3025

21

WD2_SAFETY_IN

W3024

11

WD2_COIL_N

W3023

A2

SAFETY_24V

W3022

A1

CH2_COM_+24V

CH1_COM

DRIVER_RETURN

COIL_+24V

CH2_NC_FEEDBACK

CH2_NO_UNUSED

CH1_NC_UNUSED

CH1_NO

22

W3027

WD2_NC_24V

12

W3026

SR1_S22

PROPOSED – RECEIVED POLARITY/FMEA VERIFICATION REQUIRED

NOTE 1: Required after E–stop/watchdog dropout: cause healthy –> RESET press/release –> SAFE_READY –> distinct ARM press/release.

NOTE 2: KWD contacts are in the SR1 returns; SRA1 receives the two SR1 safety outputs directly.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED

PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

Sheet: /03 Distinct ARM and watchdog eligibility/
File: 03_arm_watchdog_eligibility.kicad_sch

Title: PB HR–V0 Electrical V3 – 03

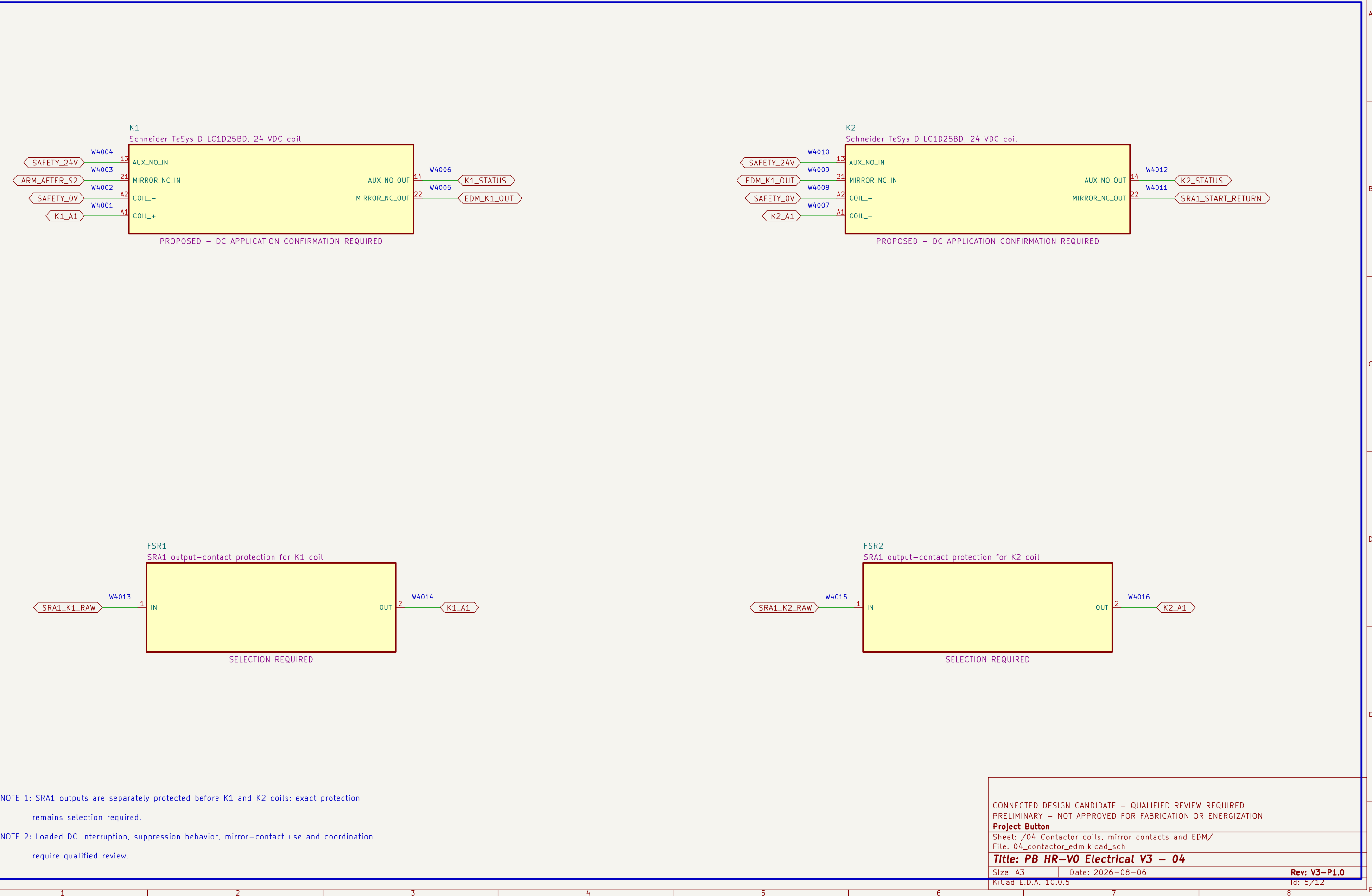
Size: A3Date: 2026–08–06

KiCad E.D.A. 10.0.5Id: 4/12

Rev: V3–P1.0

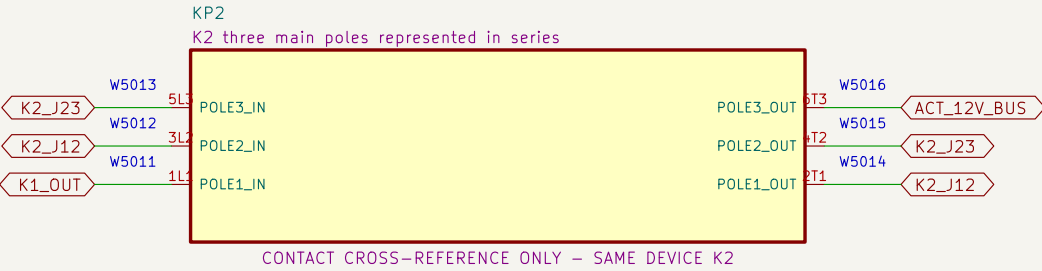
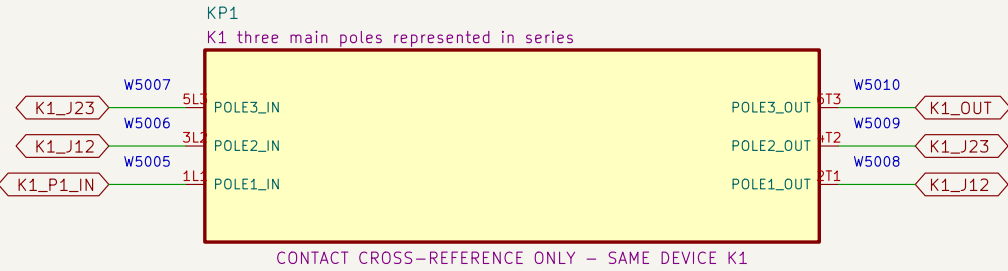
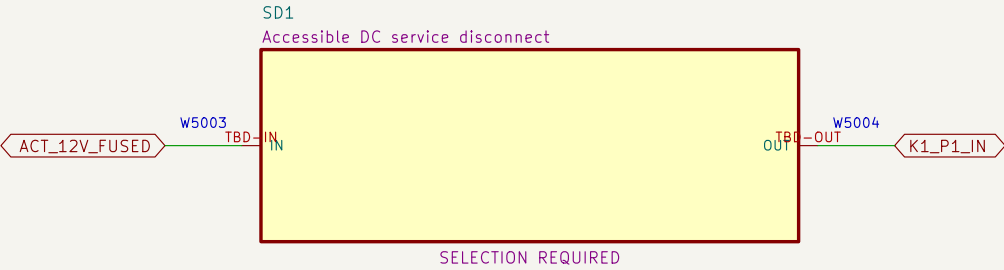
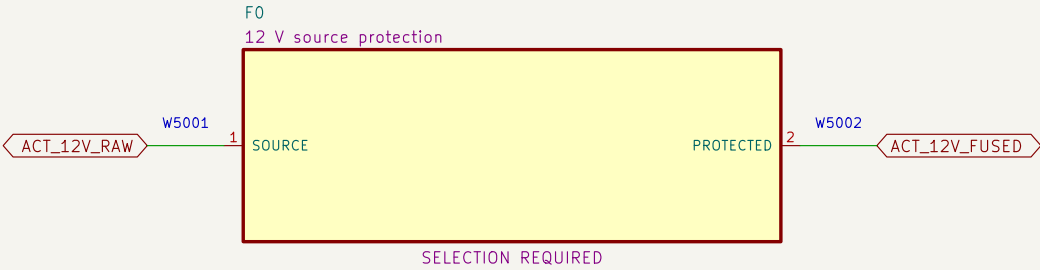
04 Contactor coils, mirror contacts and EDM

K1 and K2 are distinct final elements; their mirror contacts form the monitored restart return.



05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



NOTE 1: Series jumpers: K1 2T1→3L2, 4T2→5L3, 6T3→K2 1L1; repeat through K2 to ACT_12V_BUS.

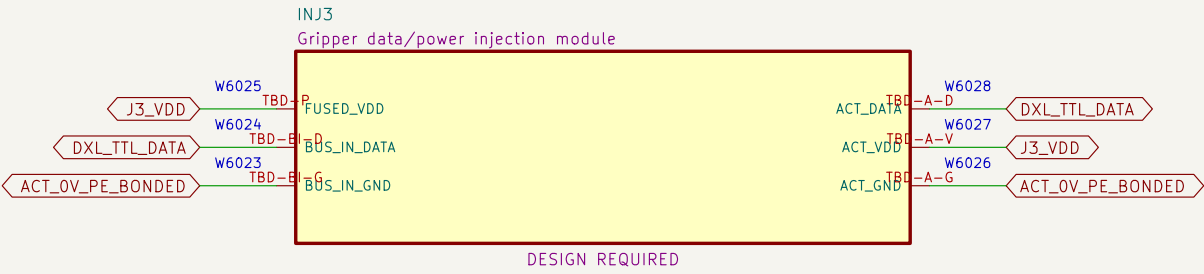
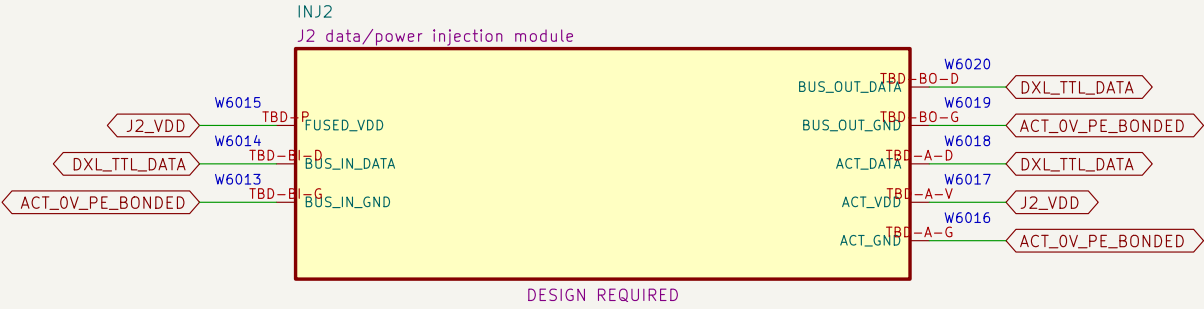
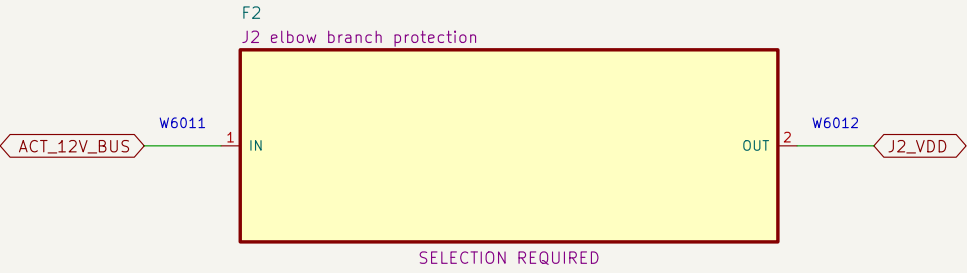
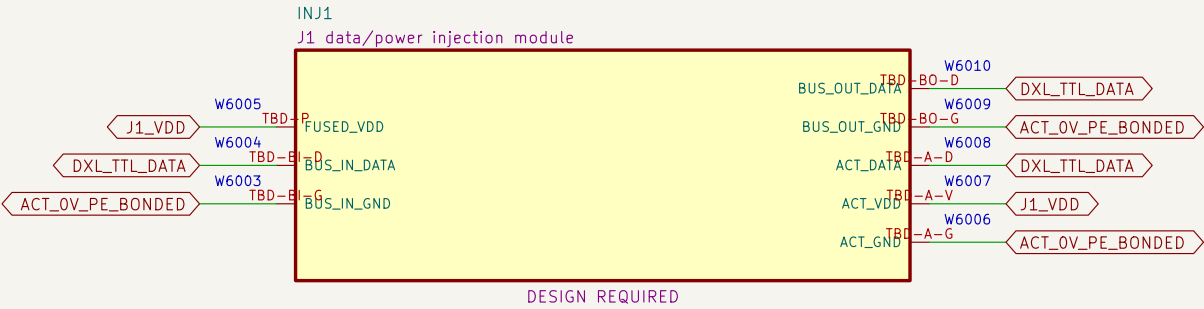
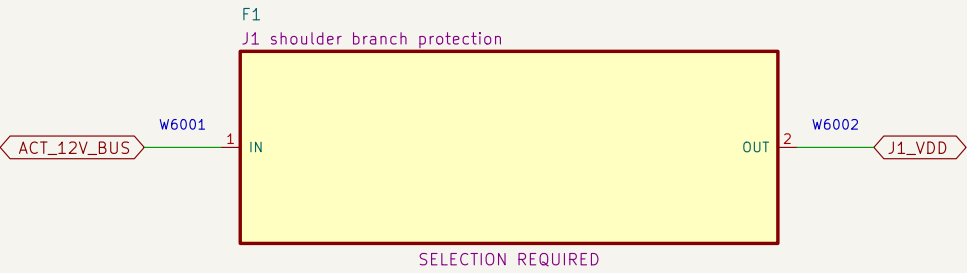
NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /05 Redundant actuator–power interruption/
File: 05_actuator_interruption.kicad_sch

Title: PB HR–V0 Electrical V3 – 05		Rev: V3–P1.0
Size: A3	Date: 2026–08–06	
KiCad E.D.A. 10.0.5		Id: 6/12

06 Protected actuator branches and VDD-isolating injection

Each actuator has a separate protected VDD branch; inter-actuator data cables must omit VDD.



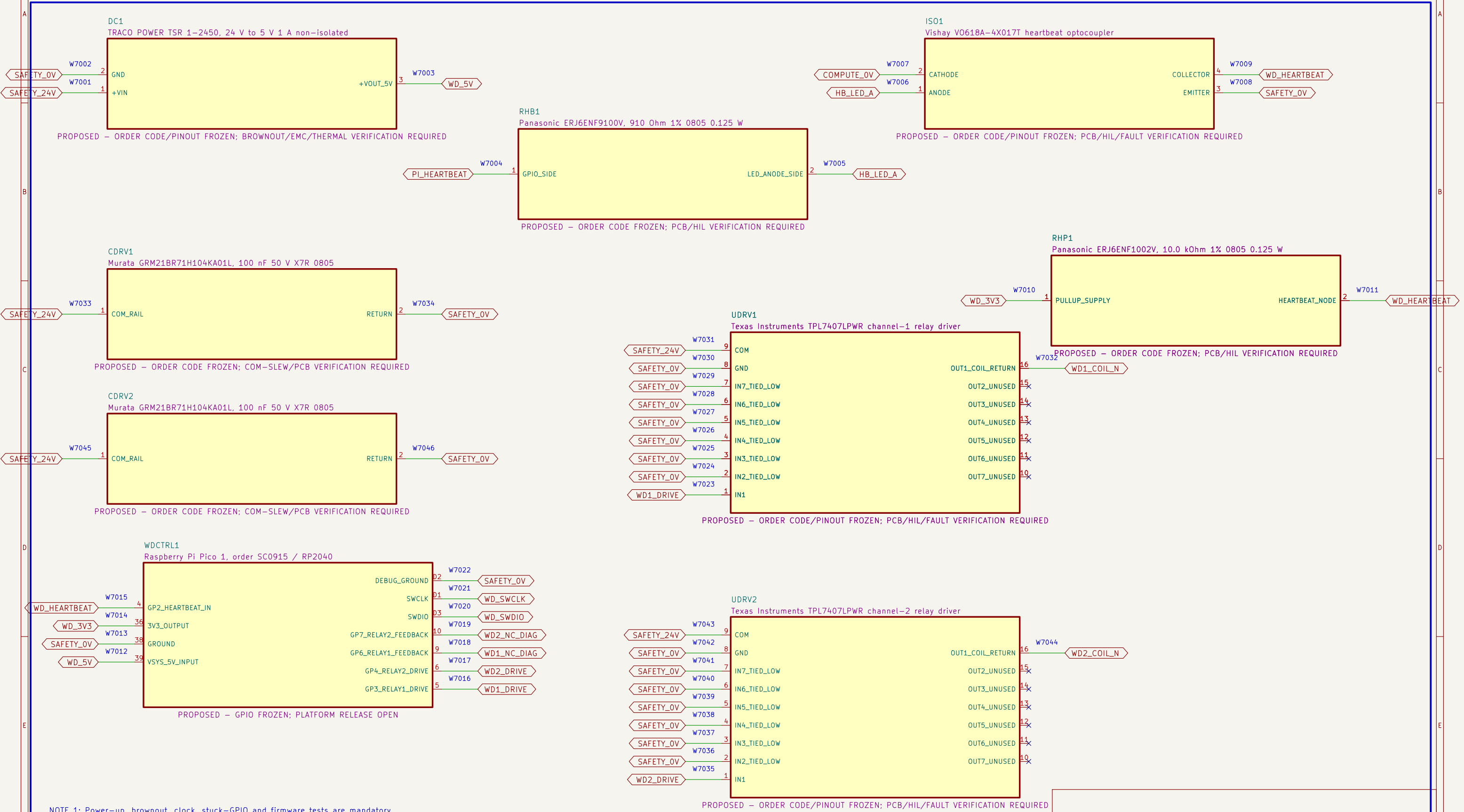
NOTE 1: Standard fully populated ROBOTIS TTL daisy-chain cables are prohibited because they would parallel branch VDD.

NOTE 2: Every injection module requires released source, continuity, isolation, pull and no-backfeed tests.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /06 Protected actuator branches and VDD-isolating injection/
File: 06_branches_and_injection.kicad_sch

07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.



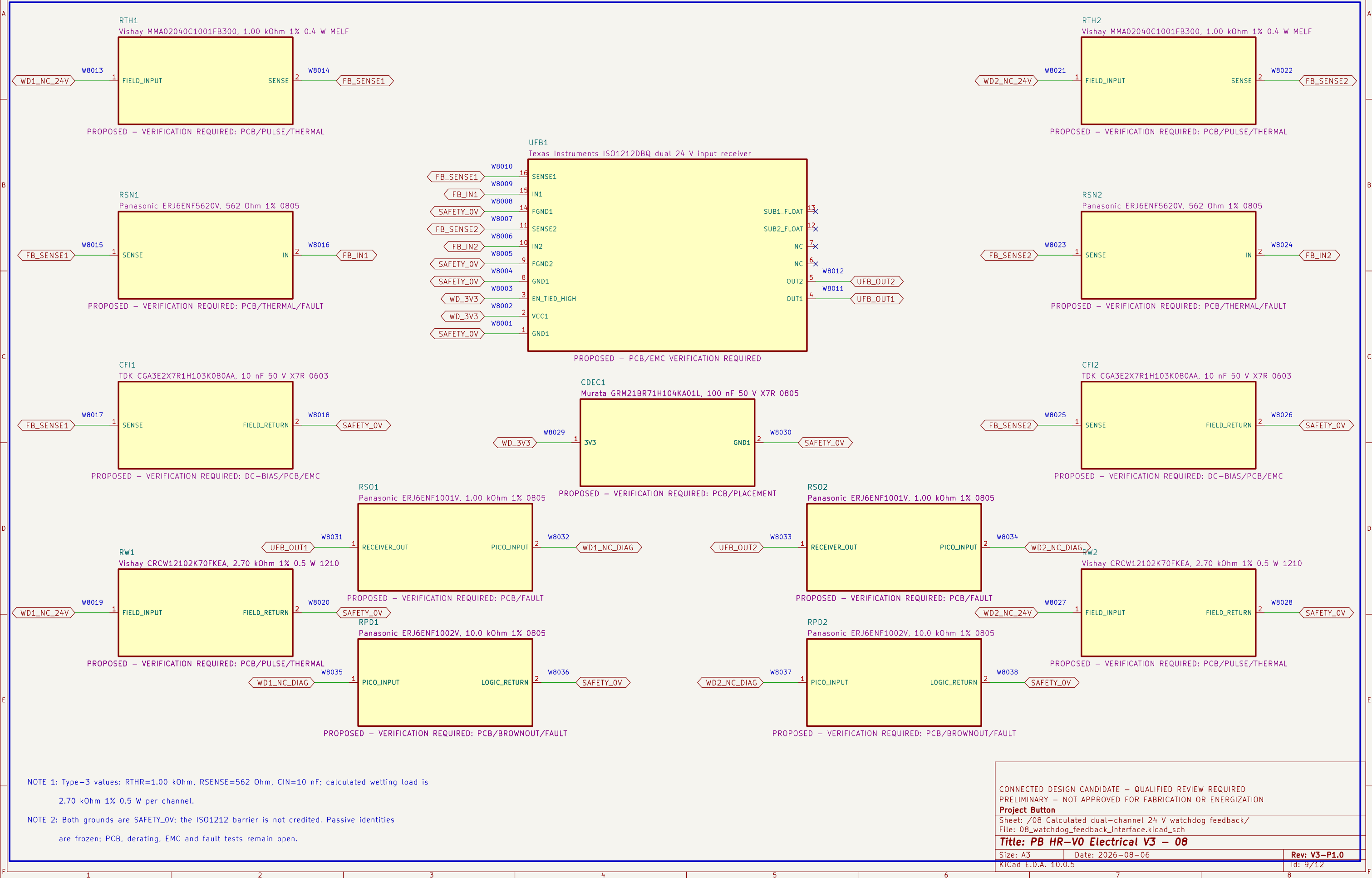
NOTE 1: Power–up, brownout, clock, stuck–GPIO and firmware tests are mandatory.

NOTE 2: Qualified review decides whether watchdog loss is credited or diagnostic only:
current safety credit is NONE.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /07 Independent watchdog power, controller and drivers/
File: 07_watchdog_control.kicad_sch

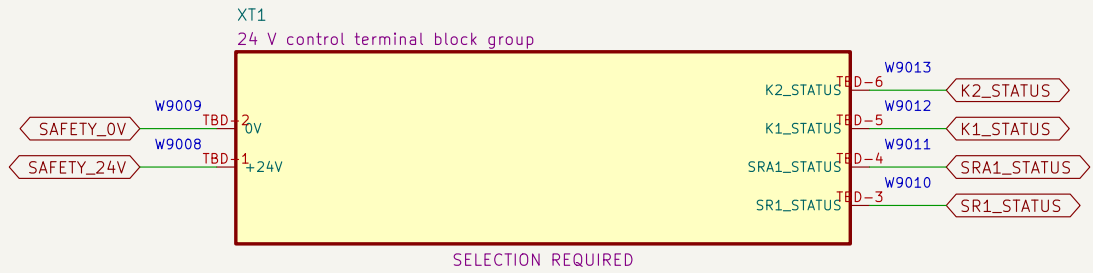
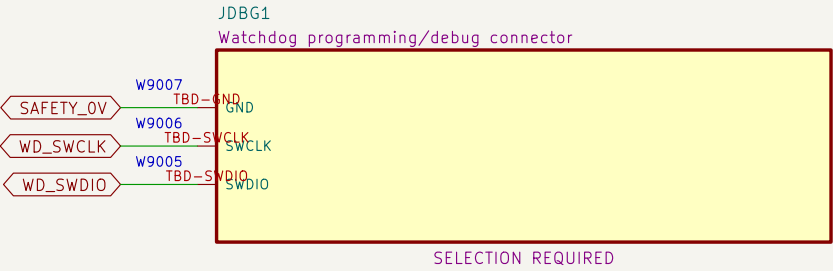
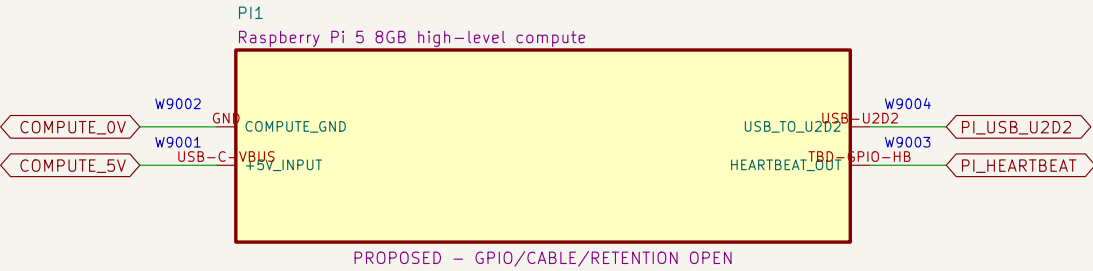
08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



09 Compute, debug and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



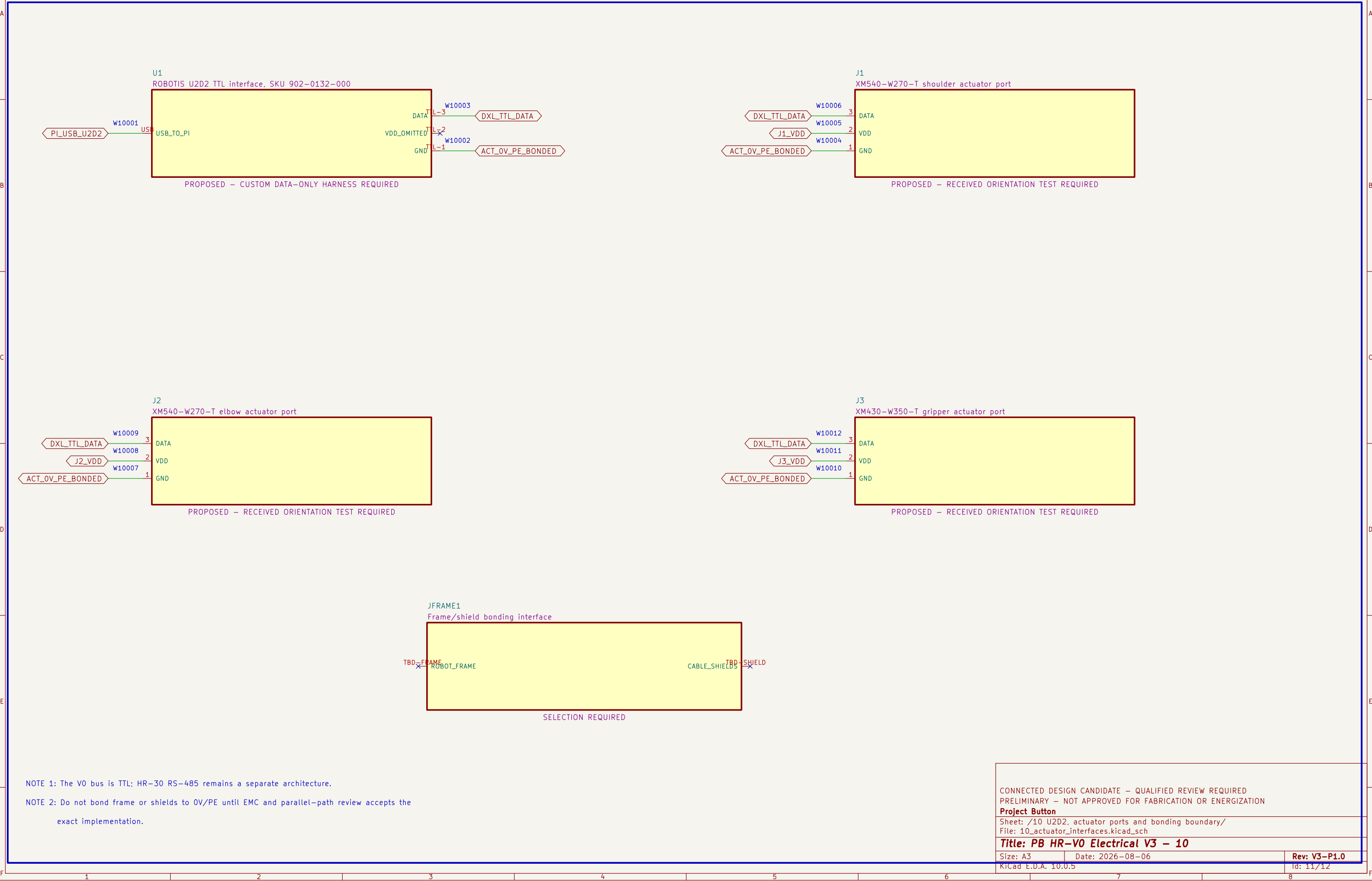
NOTE 1: Programming/debug connections must not enable outputs during operation.

NOTE 2: Terminal family, markers, conductor range, torque and enclosure layout remain
selection required.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /09 Compute, debug and control terminals/ File: 09_compute_and_control_terminals.kicad_sch		
Title: PB HR–V0 Electrical V3 – 09		
Size: A3	Date: 2026–08–06	Rev: V3–P1.0
KiCad E.D.A. 10.0.5	Id: 10/12	

10 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.



11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.

JWP1
Phoenix Contact MKDS 1/4–3,5, item 1751264

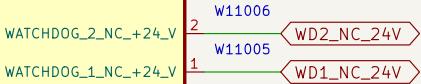


PROPOSED – VERIFICATION REQUIRED

JWF1
Phoenix Contact MKDS 1/2–3,5, item 1751248



PROPOSED – VERIFICATION REQUIRED



JWH1
Phoenix Contact MKDS 1/2–3,5, item 1751248



PROPOSED – VERIFICATION REQUIRED



NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button

Sheet: /11 Watchdog PCB external connectors/
File: 11_watchdog_pcb_connectors.kicad_sch

Title: PB HR–V0 Electrical V3 – 11

Size: A4	Date: 2026–08–06	Rev: V3–P1.0
KiCad E.D.A. 10.0.5		Id: 12/12